

Multi-Agent Scientific Paper Review

Powered by Phi-3 Mini · May 03, 2026 18:53

File	Final_paper_kjns_210426 (1).pdf
Words	12,362
Sections Detected	preamble, introduction, methodology, experiments, discussion, section_vi, conclusions, section_vii, biogra
Review Mode	FAST
Model	llama3.2

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Editorial Decision

MINOR REVISION

Weighted Score: 3.9 / 5.0
Confidence: 82%

Evaluation Score Table

Scale: 0 = Not evaluable | 1 = Very weak | 2 = Weak | 3 = Acceptable | 4 = Good | 5 = Excellent

Criterion	Score	Rating	Visual
Originality & Novelty	4.0	Good	■■■■■
Manuscript Structure	4.0	Good	■■■■■
Methodology & Equations	4.0	Good	■■■■■
Statistical Analysis	4.2	Good	■■■■■
Results & Evidence	4.2	Good	■■■■■
Discussion & Conclusions	4.2	Good	■■■■■
Figures & Tables	4.0	Good	■■■■■
Scientific Writing	3.5	Good	■■■■■
References & Citations	2.5	Weak	■■■■■
Ethics & Transparency	3.5	Good	■■■■■

Detailed Agent Reviews

TitleAbstractKeywordsReviewer

Title, Abstract, Keywords

Score: 4.0 / 5.0 | Confidence: 100%

Strengths:

- [TITLE] The title clearly signals the core method and application domain.
- [ABSTRACT] The abstract provides substantial methodological and validation detail.
- [ABSTRACT] Quantitative result signals are present in the abstract.
- [KEYWORDS] 8 keywords were detected for indexing and discoverability.

Weaknesses:

- [TITLE] could be more concise

Major Comments:

- [ABSTRACT] lacks a clear background section

StructureReviewer

Manuscript structure and logical coherence

Score: 4.0 / 5.0 | Confidence: 80%

Weaknesses:

- Introduction lacks clarity and detail
- Transitions between sections could be improved

Major Comments:

- The manuscript's structure is mostly logical, but the introduction could benefit from more context and details.
- The discussion section feels a bit disconnected from the rest of the manuscript.

Recommendations:

- Revise the introduction to provide more context and clarity
- Improve transitions between sections for better coherence

MethodologyReviewer

Methodology, experimental design, equations — holistic review

Score: 4.0 / 5.0 | Confidence: 80%

Strengths:

- The proposed approach integrates physical admissibility constraints into the neural mapping.
- A two-phase evolutionary search strategy is employed to decouple the computationally intensive training phase from deployment.

Major Comments:

- The logical sequence of the paper could be improved by explicitly stating the problem, model, method, experiment, results, and conclusion in a clear and concise manner.
- The validation directly tests what the method claims to solve, which is a strength.

Recommendations:

- Consider adding a section on potential biases and their mitigation strategies.
- Provide more details on the dataset used for experimentation and its preprocessing steps.

StatisticsReviewer

Statistical methods, tests, and reporting quality

Score: 4.2 / 5.0 | Confidence: 80%

Strengths:

- Bootstrap confidence intervals detected
- Holm correction applied correctly

Major Comments:

- The choice of Kruskal-Wallis test may not be appropriate for all data distributions. Consider using non-parametric tests or transforming the data to meet parametric assumptions.

Recommendations:

- Consider providing a power analysis or justification for the chosen sample size.
 - Verify that the Kruskal-Wallis test is suitable for the data distribution and consider alternative tests if necessary.
-

FiguresTablesEquationsReviewer

Figures, tables, equations — quality and coherence with text

Score: 4.0 / 5.0 | Confidence: 90%

Weaknesses:

- Some figures lack clear labels or unit information.
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ResultsReviewer

Results validity and empirical support

Score: 4.2 / 5.0 | Confidence: 80%

Strengths:

- The manuscript presents a clear research objective and framework for the proposed EANN approach.

Weaknesses:

- Some details about the preprocessing steps are missing, which might affect the accuracy of the model.

Major Comments:

- While the manuscript reports various quantitative results, it would be beneficial to provide more context on how these results were obtained and validated.

Recommendations:

- Provide additional details about the preprocessing steps applied to the measured signals.
 - Consider including more information about the validation process, such as the statistical methods used for comparison.
-

DiscussionConclusionsReviewer

Discussion depth and conclusions validity

Score: 4.2 / 5.0 | Confidence: 80%

Weaknesses:

- The Discussion could benefit from a more in-depth comparison with prior work on structural parameter identification methods.

Major Comments:

- The Discussion effectively highlights the advantages of EANN over GA and PSO, but could delve deeper into the theoretical implications of these findings.
- The manuscript could provide more context on the limitations of classical metaheuristic approaches and how EANN addresses them.

Recommendations:

- Consider adding a section to discuss potential applications and challenges for EANN in real-world structural identification scenarios.

- Provide more details on the experimental setup and data preprocessing steps to increase transparency and reproducibility.

WritingReviewer

Scientific writing quality, clarity, and style

Score: **3.5 / 5.0** | Confidence: 80%

Weaknesses:

- Overuse of passive voice (e.g., 'This study presents a physics-guided Evolutionary Artificial Neural Network')
- Unnecessary repetition (e.g., 'including structural health monitoring, active, semi-active, and passive vibration control for seismic response mitigation')

Major Comments:

- Consider rephrasing sentences to improve clarity and concision
- Define technical terms (e.g., EANN) on first use to ensure consistency

Recommendations:

- Use active voice instead of passive voice where possible
- Provide definitions for technical terms (e.g., EANN) on first use

ReferencesReviewer

References quality, recency, and relevance

Score: **2.5 / 5.0** | Confidence: 90%

Weaknesses:

- References are mostly from conference papers, which may not be as widely available or well-indexed as journal articles.
- Some references ([21], [22]) appear to have minor formatting issues (e.g., missing DOI, inconsistent spacing).

Major Comments:

- The manuscript could benefit from a more diverse range of reference sources, including journal articles and books.
- While the recent references are up-to-date, it would be beneficial to include some seminal works from the early 2000s or before to provide historical context.

Minor Comments:

- The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Recommendations:

- Consider adding a brief description of these references in the introduction to provide context for readers unfamiliar with the topic.
- Consider including some journal articles or books alongside conference papers to increase diversity in reference sources.

EthicsLimitationsReviewer

Research ethics, limitations, and transparency

Score: **3.5 / 5.0** | Confidence: 60%

Strengths:

- No clear human/animal-subject signal was detected in the parsed manuscript.

Weaknesses:

- Conflict-of-interest declaration was not detected.

Major Comments:

- Conflict-of-interest declaration was not detected.

Recommendations:

- Include a conflict-of-interest declaration.

Consolidated Major Comments

These issues must be addressed before the manuscript can be accepted.

- [1] [ABSTRACT] lacks a clear background section
- [2] The manuscript's structure is mostly logical, but the introduction could benefit from more context and details.
- [3] The discussion section feels a bit disconnected from the rest of the manuscript.
- [4] The logical sequence of the paper could be improved by explicitly stating the problem, model, method, experiment, results, and conclusion in a clear and concise manner.
- [5] The validation directly tests what the method claims to solve, which is a strength.
- [6] The choice of Kruskal-Wallis test may not be appropriate for all data distributions. Consider using non-parametric tests or transforming the data to meet parametric assumptions.
- [7] While the manuscript reports various quantitative results, it would be beneficial to provide more context on how these results were obtained and validated.
- [8] The Discussion effectively highlights the advantages of EANN over GA and PSO, but could delve deeper into the theoretical implications of these findings.
- [9] The manuscript could provide more context on the limitations of classical metaheuristic approaches and how EANN addresses them.
- [10] Consider rephrasing sentences to improve clarity and concision
- [11] Define technical terms (e.g., EANN) on first use to ensure consistency
- [12] The manuscript could benefit from a more diverse range of reference sources, including journal articles and books.

Consolidated Minor Comments

These are suggestions that would improve the manuscript quality.

- [1] The parser detected approximately 34 references; assess recency and relevance from the visible list rather than treating the section as empty.

Specific Improvement Recommendations

1. Revise the introduction to provide more context and clarity
 2. Improve transitions between sections for better coherence
 3. Consider adding a section on potential biases and their mitigation strategies.
 4. Provide more details on the dataset used for experimentation and its preprocessing steps.
 5. Consider providing a power analysis or justification for the chosen sample size.
 6. Verify that the Kruskal-Wallis test is suitable for the data distribution and consider alternative tests if necessary.
 7. Provide additional details about the preprocessing steps applied to the measured signals.
 8. Consider including more information about the validation process, such as the statistical methods used for comparison.
 9. Consider adding a section to discuss potential applications and challenges for EANN in real-world structural identification scenarios.
 10. Provide more details on the experimental setup and data preprocessing steps to increase transparency and reproducibility.
 11. Use active voice instead of passive voice where possible
 12. Provide definitions for technical terms (e.g., EANN) on first use
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